NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-83446/8D
6 July 1993
SUPERSEDING
MIL-C-83446/8C
25 February 1988

PERFORMANCE SPECIFICATION SHEET

COILS, RADIO FREQUENCY, CHIP, FIXED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83446.

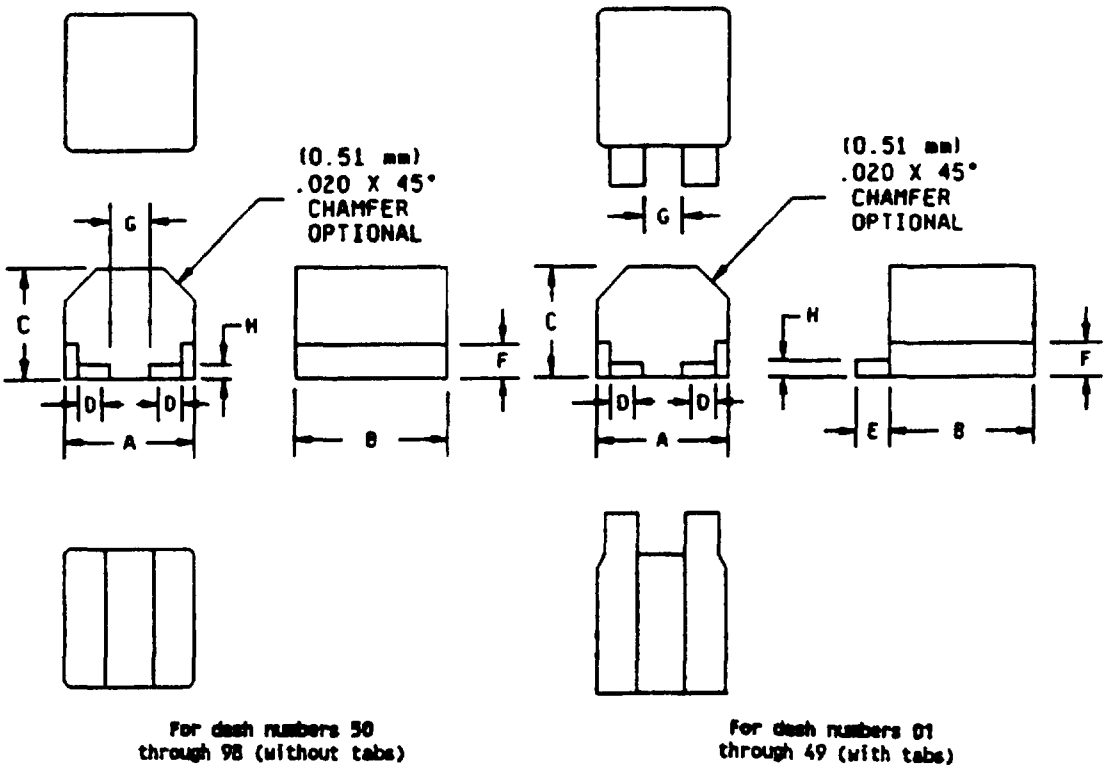


FIGURE 1. Fixed, chip coil.

(D) denotes changes.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited

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For dash numbers 01 through 49 (with tabs)		
Ltr	Inches	mm
A	.110 max	2.79 max
B	.105 max	2.67 max
C	.070 max	1.78 max
D	.030 ±.005	0.76 ±0.13
E	.040 +.015 −.010	1.02 +0.38 −0.25
F	.018 ±.005	0.46 ±0.13
G	.038 ±.010	0.97 ±0.25
H	.005 ±.002	0.13 ±0.05

For dash numbers 50 through 98 (without tabs)		
Ltr	Inches	mm
A	.110 max	2.79 max
B	.105 max	2.67 max
C	.070 max	1.78 max
D	.030 ±.005	0.76 ±0.13
F	.018 ±.005	0.46 ±0.13
G	.038 ±.010	0.97 ±0.25
H	.005 ±.002	0.13 ±0.05

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Fixed, chip coil - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Weight: 0.5 gram maximum.

Operating temperature range: -55°C to +125°C.

Temperature rise (at 90°C): 35°C.

Maximum operating temperature: 125°C.

Altitude: 70,000 feet.

Dielectric withstanding voltage: Method 301 of MIL-STD-202, test voltage 300 volts rms.

Barometric pressure: Method 105, test condition C, MIL-STD-202 (70,000 feet), test voltage 200 volts rms.

Electrical characteristics (initial): See table I.

Electrical characteristics (final): See table II.

Supersession data: A new part numbering system with code for termination materials was incorporated with revision A, superseding MIL-I-83446/8, dated 1 March 1979.

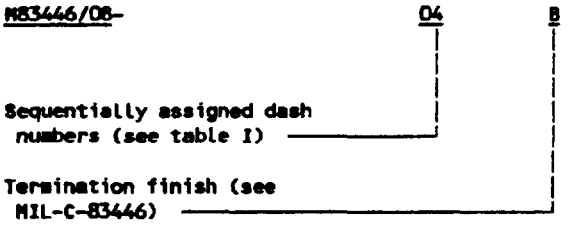
M83446/08-(dash number from table I) supersedes M83446/8-(dash number from table I).

Examples:

M83446/08-21C supersedes M83446/8-21.

M83446/08-03C supersedes M83446/8-3.

(D) Part or Identifying Number (PIN): The PIN shall be in the following form:



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TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number	Inductance ±10% (µH)	Q (min) Ⓓ	Q (typ) Ⓓ	Test frequency (MHz) 1/	Self-resonant frequency (min) (MHz)	DC resistance (max) (Ohms)	Current (max) (mA) 2/
01, 50	0.010	50	55	150	2000	0.025	750
02, 51	0.012	50	55	150	2000	0.025	750
03, 52	0.015	50	55	150	1800	0.040	750
04, 53	0.018	50	55	150	1500	0.040	750
05, 54	0.022	45	50	100	1400	0.040	750
06, 55	0.027	45	50	100	1200	0.040	750
07, 56	0.033	47	55	100	1100	0.050	640
08, 57	0.039	47	55	100	1000	0.07	600
09, 58	0.047	47	55	100	900	0.08	550
10, 59	0.056	47	55	100	850	0.09	520
11, 60	0.068	47	55	100	840	0.10	480
12, 61	0.082	47	55	100	750	0.11	470
13, 62	0.100	47	55	50	580	0.11	470
14, 63	0.120	47	55	50	240	0.11	470
15, 64	0.150	47	55	50	230	0.12	450
16, 65	0.180	51	60	50	230	0.14	430
17, 66	0.220	51	60	50	230	0.20	350
18, 67	0.270	51	60	50	230	0.25	310
19, 68	0.330	51	60	50	200	0.30	280
20, 69	0.390	47	55	50	190	0.45	240
21, 70	0.470	47	55	25	180	0.50	230
22, 71	0.560	45	53	25	170	0.50	230
23, 72	0.680	45	53	25	160	0.50	230
24, 73	0.820	45	53	25	150	0.50	230
25, 74	1.00	45	53	25	130	0.50	230
26, 75	1.20	36	43	7.90	120	0.60	200
27, 76	1.50	36	43	7.90	100	1.1	160
28, 77	1.80	38	45	7.90	90	1.1	160
29, 78	2.20	38	45	7.90	85	1.5	130

See footnotes at end of table.

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TABLE I. Electrical characteristics (initial) and dash numbers - Continued.

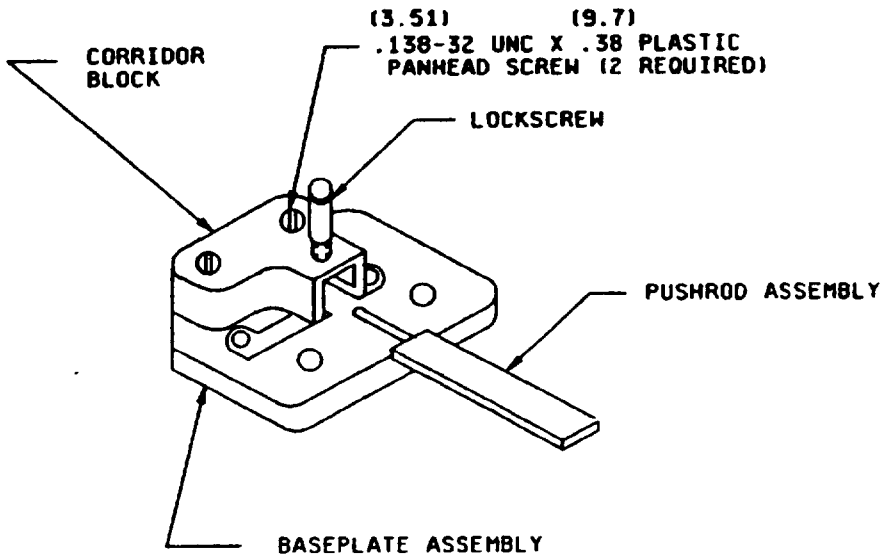
Dash number	Inductance ±10% (µH)	Q (min)	Q (typ)	Test frequency (MHz) 1/	Self-resonant frequency (min) (MHz)	DC resistance (max) (Ohms)	Current (max) (mA) 2/
30, 79	2.70	41	48	7.90	80	1.7	125
31, 80	3.30	42	50	7.90	75	1.8	120
32, 81	3.90	42	50	7.90	65	2.0	110
33, 82	4.70	41	48	7.90	55	2.3	100
34, 83	5.60	41	48	7.90	45	2.6	98
35, 84	6.80	36	43	7.90	40	2.8	94
36, 85	8.20	36	43	7.90	35	3.0	90
37, 86	10.0	36	43	7.90	33	3.3	87
38, 87	12.0	36	43	7.90	26	4.0	79
39, 88	15.0	32	38	2.50	24	4.2	79
40, 89	18.0	32	38	2.50	21	4.4	75
41, 90	22.0	32	38	2.50	19	7.5	57
42, 91	27.0	32	38	2.50	14	8.0	55
43, 92	33.0	30	35	2.50	12	13.0	45
44, 93	39.0	30	35	2.50	10	17.0	38
45, 94	47.0	30	35	2.50	9.0	19.0	36
46, 95	56.0	30	35	2.50	8.5	23.0	33
47, 96	68.0	30	35	2.50	8.2	25.0	32
48, 97	82.0	30	35	2.50	8.0	28.0	30
49, 98	100.0	30	35	2.50	7.0	31.0	28

1/ Test frequency range 0.25 through 25 MHz: For electrical characteristics measurements, use the TF-260Q-1 test fixture, or equal (see figure 3). Fixture inductance (approximately 0.028 µH) and residual Q-meter inductance (approximately 0.01 µH) should be subtracted from indicated inductance.

Test frequency range 25 through 150 MHz: For electrical characteristics measurements, use the TF-250RX-1 test fixture, or equal (see figure 2). Fixture inductance (approximately 0.009 µH) should be subtracted from calculated inductance.

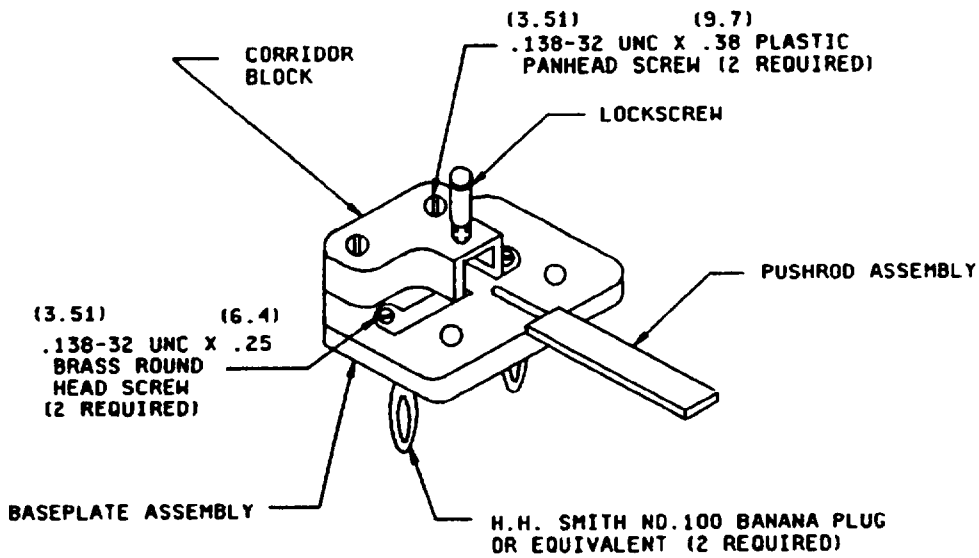
2/ Maximum current allowed is not to exceed the specified temperature rise.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 2. Typical chip coil test fixture (TF-250RX-1, or equivalent).



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 3. Typical chip coil test fixture (TF-260Q-1, or equivalent).

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TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from the initial measurements			
Qualification inspection	Inductance	DC resistance	Self-resonant frequency	Q
	Percent		Percent	Percent
Group II	±5	±(3% +0.001 ohm)	-8	-10
Group IV	±5	±(2% +0.001 ohm)	-10	-10
Group V	±2	- - - - -	- - -	-10
Quality conformance inspection				
Group C				
Subgroup II	±5	±(3% +0.001 ohm)	-8	-10
Subgroup IV	±5	±(3% +0.001 ohm)	-8	-10

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - MI
Navy - AS
Air Force - 19

User activity:
Navy - OS

Preparing activity:
Army - ER

Agent:
DLA - ES

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